



A Blockchain-Driven System for Authenticity, Sustainability and Export Readiness of Ayurvedic Herbs.

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Context of the document

This document contains a detailed view on the introduction of the project topic, has the objectives and the vision of the project listed along with hardware and software specifications. Project deliverables are listed after the outlined literature survey . Work plan (detailed implementation strategy is still in progress, however overall plan has been mentioned in the document).

Objectives

The main goal of this project is to build a robust, tech driven system that ensures the authenticity, sustainability, and export readiness of Ayurvedic herbs.

- Development of an end to end interoperable traceability ledger and QR: To utilize Hyperledger Fabric for creating an immutable record of every step in the herbal supply chain from raw material sourcing to the final consumer.
(seedling—harvest—process—distribution—export). Enable each batch's transparency using QR code scan and verification
 - Implementation of an AI powered botanical authentication: To integrate a Sequential CNN or architectures like EfficientV2M to classify medicinal leaves with an accuracy of 90% to 98%, thereby preventing misidentification and adulteration.
 - Establish origin verification through geotagging and IoT: To automate the recording of key factors such as soil quality, temperature, humidity, and geographic location directly from the seedling stage using IoT sensors to prevent source falsification thus ensuring the quality and validity of the herb.
 - Enhance sustainability monitoring: To use blockchain for documenting compliance with Good Agricultural Practices (GAP) and organic standards, providing verifiable evidence of sustainable farming for international buyers with inclusion of SDGs.
 - Achieve export readiness through international compliance: To design the data architecture to automatically verify and document adherence to international standards such as the EU-FMD and the DSCSA.
 - Optimize performance and scalability: To use Hyperledger Caliper to benchmark the blockchain network's throughput and latency, ensuring it can handle high transaction volumes from thousands of Ayush manufacturing units
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Our Vision

Our vision is to ensure an end to end model which will guarantee:-

- Botanical authenticity
 - Supply-chain transparency
 - Sustainable cultivation
 - Export-readiness compliance
 - Interoperability with government platforms
 - Inclusion of small-scale farmers and SMEs
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Literature survey results and review

Existing Work

Significant work has already been carried out in the areas of blockchain traceability, AI-based herb identification, digital governance, and IoT-enabled monitoring within the Ayurvedic ecosystem.

- Established blockchain systems such as the Indian Herbal Blockchain Network have demonstrated how a permissioned framework built on Hyperledger Fabric can record supply chain events from harvest to retail. Platforms like HerBChain have extended this model by tracking multiple lifecycle stages including plantation, processing, laboratory validation, distribution, and retail. Research has also explored double-blockchain architectures that combine private internal chains with consortium networks for public verification. In addition, segmented encryption models using AES and RSA have been proposed to protect enterprise intellectual property while still enabling regulators to verify authenticity hashes.
- In the field of artificial intelligence, extensive comparative research has been conducted on convolutional neural network architectures for medicinal plant identification. Models such as ResNet-50, VGG-16, and EfficientV2M have demonstrated classification accuracies reaching up to 98 percent under controlled conditions. Some platforms have

already integrated sequential CNN models for real-time herb verification at the point of harvest. Conversational AI systems using artificial neural networks and natural language processing have also been developed to provide Ayurvedic guidance, indicating that digital intelligence is steadily entering traditional medicine systems.

- At the government infrastructure level, digital portals such as e-Aushadhi, e-Charak, Ayush Suraksha, and the Traditional Knowledge Digital Library have digitized licensing, pharmacovigilance, sourcing information, and classical formulation archives. These initiatives have laid a strong administrative and documentation foundation for modernization.
- IoT-based models have further introduced automated environmental recording from the seedling stage, including soil quality, humidity, and temperature monitoring. Geo-tagging modules integrated with wireless transmission systems have been conceptualized to send real-time harvesting events to distributed ledgers. Performance benchmarking studies using Hyperledger Caliper have demonstrated that enterprise blockchain networks can sustain stable throughput with acceptable latency under moderate transaction loads.

Together, these contributions form a substantial technological base. However, they do not yet constitute a unified, interoperable, and export-ready ecosystem.

What Is Still Missing (The Gaps)

Despite strong foundational research, several structural gaps remain unresolved.

- First, interoperability remains limited. Digital systems such as e-Aushadhi and e-Charak operate independently, and there is no unified data layer linking them to a blockchain ledger. Without integration, traceability remains fragmented rather than continuous.
- Second, export compliance automation is largely absent. While domestic regulations exist, there is no automated mapping between Indian compliance records and international mandates such as the European Union Falsified Medicines Directive or the US Drug Supply Chain Security Act. This gap restricts global scalability.
- Third, most scalability studies are conducted in small network environments. There is insufficient empirical testing of how blockchain systems perform when scaled to represent thousands of Ayush manufacturing units or when transactions originate from remote rural regions.
- Fourth, sustainability tracking remains conceptual. Real-time carbon footprint estimation, biodiversity indicators, and measurable Good Agricultural Practice compliance are rarely embedded into blockchain systems in a verifiable manner.
- Fifth, the digital divide among micro-scale farmers remains inadequately addressed. Many stakeholders lack stable internet connectivity or advanced digital literacy. Without simplified and offline-capable interfaces, technological inclusion remains limited.

- Sixth, the integration of digital forensics with classical Ayurvedic manuscripts is underdeveloped. Modern supply chain records are not systematically linked to authenticated textual sources, which weakens historical validation in academic and regulatory contexts.
 - Seventh, standardized data models for herbal traceability are still missing. Inconsistent data schemas across institutions limit interoperability and advanced analytics.
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Scope and deliverables

Project Deliverables

- A detailed system architecture showing the interoperable blockchain-based Ayurvedic herb supply chain, including the roles of stakeholders, the flow of data, and the ways to control access.
- Implement export-compliance auto-verification architecture.
- Create a sustainability scoring model (“Green Rating” per batch).
- Introduce forensic timestamping for authenticity validation.
- A permissioned blockchain network with separate nodes for farmers, processors, exporters, certification bodies, and regulatory bodies.
- Smart contracts were made to handle things like registering herbs, making batches, keeping records of harvesting, processing details, quality testing, uploading certifications, and giving permission to export.
- A database of Ayurvedic herbs that is linked to a blockchain. It has botanical names, Ayurvedic classifications, geographical origins, batch IDs, sustainability indicators, and compliance records.
- A verification system based on QR codes that lets consumers, exporters, and government agencies quickly access information that has been verified by the blockchain.
- A web-based or app-based user interface that gives authorized users access to dashboards and end users access to a read-only verification interface.
- A report on the system's security and integrity that looks at how well it can stop data tampering, fraud, and unauthorized access.
- A case study showing how the proposed blockchain system can track selected Ayurvedic herbs like Ashwagandha, Turmeric, or Tulsi from start to finish.
- Results of a performance evaluation that looked at the blockchain network's transaction latency, scalability, and storage efficiency.
- All the project paperwork, such as the final report, an explanation of the system design, the results of the experiments, the future scope, and the materials for the presentation.

SDG covered

Primary:

- SDG 3 - Good health and well being
- SDG 12 - Responsible consumption and production
- SDG 15 - Life on land

Secondary:

- SDG 17 - Partnership for the Goals
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Hardware and Software requirements

Hardware Requirements

Component	Specifications	Purpose
Edge Computing Node	Raspberry Pi 4 (4GB RAM)	Serves as the local blockchain peer and API server.
Storage	32GB MicroSD (Class 10)	Hosts the Ubuntu OS, Docker containers, and local logs.

Connectivity	A9G GPS + GSM Module	Captures geo-coordinates and transmits collection events from remote sites via GSM.
Imaging Hardware	Raspberry Pi Camera Module	(New) Captures high-resolution images of medicinal leaves for AI-based botanical authentication.
Environmental Sensors	DHT22 (Temp/Humidity) & Soil Moisture Sensors	(New) Records "ground truth" data (soil, temperature, moisture) from the seedling stage to ensure herb quality and sustainability.
Power Supply	10,000 mAh Power Bank	Provides portable power for field-based data collection.
Batch Identification	QR Code Label Printer	Generates physical labels for batches that link to the Digital Product Passport.
Local Network	Internet Router / Broadband	Facilitates communication between the edge node and the wider blockchain consortium.

Software Requirements

Component	Technology	Purpose
Blockchain Framework	Hyperledger Fabric	Permissioned ledger for secure, private supply-chain transactions.

Containerization	Docker & Docker Compose	Manages the deployment of blockchain nodes and microservices.
Backend Environment	Node.js & Express.js	Handles REST APIs and chaincode execution for transactions.
AI/ML Framework	TensorFlow / Keras	(New) Runs the Sequential CNN models (e.g., ResNet-50 or EfficientV2M) for leaf classification with up to 98% accuracy.
Decentralized Storage	IPFS (InterPlanetary File System)	(New) Securely stores high-resolution herb images off-chain to prevent blockchain bloat; only the hash is recorded on the ledger.
Databases	PostgreSQL & MongoDB	(Enhanced) Use PostgreSQL for structured pharmacopoeial standards and MongoDB for unstructured analytics/dashboards.
Frontend Framework	React.js (as a PWA)	Provides a mobile-friendly interface with offline capabilities for rural stakeholders.
Mapping Utility	Leaflet.js	Displays interactive maps of geo-tagged herb origin locations.
Performance Testing	Hyperledger Caliper	(New) Benchmarks throughput and latency to ensure the system can handle thousands of manufacturing units for export readiness.
Operating System	Ubuntu Server 20.04 LTS	The stable underlying environment for all system components.

Work Plan/Timeline (implementation plan)

March 2026 Hardware components layout design and soldering	April 2026 Soldering of components and designing software	May-June Working and building CNN, ML algos to train models, Website	July Software - Hardware integration and testing.	August Final touches and finishing the final model.
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Next steps

- ☐ Design Software architecture
- ☐ Design PCB Layout
- ☐ Graphical abstract /Concept diagram
- ☐ Circuit diagram and UML diagrams
- ☐ PCB soldering
- ☐ Soldering Components
- ☐ Design Casing
- ☐ Working on the software
- ☐ Designing of Website to host the entire program
- ☐ Designing the blockchain layer and CNN layer
- ☐ Using ML algorithms

Research paper links

https://drive.google.com/file/d/1b6y2Y6igI-wEJcWNTyW_E9dn1s_c7SZ/view?usp=drive_link

https://drive.google.com/file/d/1_3FnpM3GXRsZftnrcQ4OiiYdReAvA0OF/view?usp=drive_link

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